Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Database System Lab

Project: Chat System

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Introduction:

The purpose of the chat system is to facilitate communication and interaction between users through text, multimedia messages, and group chats. The system allows users to exchange messages, create and participate in group chats, send attachments (such as images, videos, and files), and receive notifications about various events, such as new messages, calls, or group invitations.

Database Design and Implementation:

1. **Establishing Tables and Adding Data:** We'll create database tables to store information of chat system and insert sample data using SQL commands.

2. **Enforcing Attribute Constraints:** We'll use constraints like Unique, NOT NULL, NULL and DEFAULT to ensure data integrity and validity.

3. **Implementing Key and Referential Integrity:** Primary and Foreign Key constraints are used to maintain data consistency and define relationships between tables.

4. **Attribute Constraints:**

- **NOT NULL:** Ensures that a column cannot have a NULL value.
- **NULL:** NULL is useful when you want to indicate that the value for a particular column is unknown or not applicable.
- **UNIQUE:** UNIQUE is applied to prevent duplicate entries in a column, which is important for fields that must have unique values.

5. **Primary Key:**

- Primary key always has unique data
- A primary key cannot have null value.
- A table can contain only one primary key constraint

6. **Foreign Key:**

- FOREIGN KEY is a key used to link two tables together.
- A FOREIGN key in one table used to point PRIMARY key in another table.

7. **INSERT INTO:** Add new rows of data into a specified table in a database.

8. **UPDATE:** Modify existing rows of data in a specified table based on certain conditions.

9. **ALTER:** Modify the structure of an existing database table, such as adding, deleting, or modifying columns.

10. **JOINS:**
    - **INNER JOIN:** It retrieves records that have matching values in both tables involved in the join.
    - **LEFT JOIN:** It returns all records from the left table and the matched records from the right table.
    - **RIGHT JOIN:** It returns all records from the right table and the matched records from the left table.
    - **CROSS JOIN:** It returns the Cartesian product of the two tables, meaning it combines each row from the first table with every row from the second table, resulting in all possible combinations.

11. **DISTINCT:** Removes duplicate records from the result set, returning only unique values.

12. **LIKE Operator:** Used to search for a specified pattern in a column.

13. **NOT LIKE Operator:** Used to exclude rows that match a specified pattern.

14. **IN Operator:** Find rows where a column's value matches any value in a list.

15. **BETWEEN Operator:** Used to find rows with values within a specific range.

16. **NOT BETWEEN Operator:** Used to find rows with values outside a specific range.

17. **Logical Operators (AND, OR):** Combining conditions in queries.

18. **ORDER BY:** Sort the results by one or more columns.

19. **Sub-Queries:** Enclosed within parentheses, used within another query for filtering, joining, or calculations.

20. **EXISTS Function:** Shows existence of rows returned by a subquery.

21. **Aggregate Functions:**

- **COUNT(*):** Returns the total number of rows in a table, including rows with NULL values.

- **SUM:** Calculates the total sum of a numeric column.
- **AVG:** Returns the average value of a numeric column.
- **MIN:** Finds the smallest value in a column.
- **MAX:** Finds the largest value in a column.

22. **GROUP BY Clause:** The GROUP BY clause is used in conjunction with the SELECT statement and Aggregate functions to group rows together by common column values.

23. **HAVING Clause:** Filters grouped rows based on specified conditions after grouping and aggregate calculations.

Conclusion:

In conclusion, the chat system provides a comprehensive platform for seamless communication and interaction among users. With features like text and multimedia messaging, group chat creation, and attachment sharing, it promotes effective collaboration and connectivity. Moreover, real-time notifications ensure users stay updated on important events, enhancing their overall experience. Overall, the project aims to foster efficient and engaging communication while catering to diverse user needs in a modern digital environment.

INSERT

```
INSERT INTO chatsystem.users (userid, username, statusmessage, lastseen, email, contact)
VALUES
(1, 'Mahrosh', 'Feeling dizzy', '2024-04-28 10:30', 'mh@gmail.com', 98765),
(2, 'Haider', 'Feeling blessed', '2002-07-14 06:09', 'haa@gmail.com', 43210),
(3, 'Asfar', 'Feeling blissful', '2019-09-05 12:40', 'er5@gmail.com', 54862),
(4, 'Sheza', 'Feeling relaxed', '2023-07-06 10:12', 'lak@gmail.com', 73158);
SELECT * FROM chatsystem.users;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

userid	username	statusmessage	lastseen	email	contact
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	54862
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	73158
NULL	NULL	NULL	NULL	NULL	NULL

UPDATE

```
UPDATE chatsystem.users
SET contact = '0334509'
WHERE userid = 3;
SELECT * FROM chatsystem.users;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

userid	username	statusmessage	lastseen	email	contact
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	73158
NULL	NULL	NULL	NULL	NULL	NULL

ALTER

```
ALTER TABLE chatsystem.users
ADD Passwords varchar(255)
AFTER contact;
SELECT * FROM chatsystem.users;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

userid	username	statusmessage	lastseen	email	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

DISTINCT

```
SELECT DISTINCT username FROM users;
```

Result Grid | Filter Rows: | Export:

username
Mahrosh
Haider
Asfar
Sheza

INNER JOIN

```
SELECT u.userid, u.username, g.groupname, g.grouppermission, g.creationdate
FROM chatsystem.users u
INNER JOIN chatsystem.groupchat g
ON u.userid = g.admins
WHERE g.grouppermission = 'private'
GROUP BY u.userid, u.username, g.groupname, g.grouppermission, g.creationdate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid	username	groupname	grouppermission	creationdate
1	Mahrosh	Besties	private	2012-12-01
3	Asfar	Forever	private	2018-06-11

LEFT JOIN

```
SELECT u.userid, u.username, g.groupname, g.grouppermission, g.creationdate
FROM chatsystem.users u
LEFT JOIN chatsystem.groupchat g
ON u.userid = g.admins
WHERE g.grouppermission = 'private'
GROUP BY  u.userid, u.username, g.groupname, g.grouppermission, g.creationdate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid	username	groupname	grouppermission	creationdate
1	Mahrosh	Besties	private	2012-12-01
3	Asfar	Forever	private	2018-06-11

RIGHT JOIN

```
SELECT u.userid, u.username, g.groupname, g.grouppermission, g.creationdate
FROM chatsystem.users u
RIGHT JOIN chatsystem.groupchat g
ON u.userid = g.admins
WHERE g.grouppermission = 'public'
GROUP BY u.userid, u.username, g.groupname, g.grouppermission, g.creationdate;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid	username	groupname	grouppermission	creationdate
2	Haider	PAFLA	public	2022-05-02
4	Sheza	PSL	public	2002-05-07

CROSS JOIN

```
SELECT *
FROM chatsystem.users u
CROSS JOIN chatsystem.groupchat g;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid	username	statusmessage	lastseen	email	contact	Passwords	groupid	groupname	grouppermission	admins	creationdate	membercount	chatlock
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL	4	PSL	public	4	2002-05-07	894	not locked
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL	3	Forever	private	3	2018-06-11	19	locked
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL	2	PAFLA	public	2	2022-05-02	6994	not locked
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL	1	Besties	private	1	2012-12-01	5	locked
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL	4	PSL	public	4	2002-05-07	894	not locked
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL	3	Forever	private	3	2018-06-11	19	locked
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL	2	PAFLA	public	2	2022-05-02	6994	not locked
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL	1	Besties	private	1	2012-12-01	5	locked
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL	4	PSL	public	4	2002-05-07	894	not locked
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL	3	Forever	private	3	2018-06-11	19	locked
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL	2	PAFLA	public	2	2022-05-02	6994	not locked
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL	1	Besties	private	1	2012-12-01	5	locked

Result 11

LIKE OPERATOR

```
SELECT * FROM chatsystem.users
WHERE username LIKE 'A%';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

userid	username	statusmessage	lastseen	email	contact	Passwords
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

NOT LIKE OPERATOR

```
SELECT * FROM chatsystem.users
WHERE username NOT LIKE 'A%';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

userid	username	statusmessage	lastseen	email	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

BETWEEN OPERATOR

```
SELECT * FROM chatsystem.users
WHERE userid BETWEEN 1 AND 3;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

userid	username	statusmessage	lastseen	email	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	0334509	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

NOT BETWEEN OPERATOR

```
SELECT * FROM chatsystem.users
WHERE userid NOT BETWEEN 3 AND 4;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

userid	username	statusmessage	lastseen	email	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	98765	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	43210	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

AND OPERATOR

```
SELECT * FROM chatsystem.users
WHERE age>=16 AND age<=45;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

userid	username	statusmessage	lastseen	email	age	contact	Passwor
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

OR OPERATOR

```
SELECT * FROM chatsystem.users
WHERE age>=16 OR userid>=2;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

userid	username	statusmessage	lastseen	email	age	contact	Passwords
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

IN OPERATOR

```
SELECT * FROM chatsystem.users
WHERE age IN(10,29);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

userid	username	statusmessage	lastseen	email	age	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	10	98765	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

NOT IN OPERATOR

```
SELECT * FROM chatsystem.users
WHERE age NOT IN(10,29);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

userid	username	statusmessage	lastseen	email	age	contact	Passwords
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ORDER BY FUNCTION

```
SELECT * FROM chatsystem.users
ORDER BY userid;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Ca

userid	username	statusmessage	lastseen	email	age	contact	Passwords
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	10	98765	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509	NULL
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ORDER BY FUNCTION (DESC)

```
SELECT * FROM chatsystem.users
ORDER BY userid DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

userid	username	statusmessage	lastseen	email	age	contact	Passwords
4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158	NULL
3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509	NULL
2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210	NULL
1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	10	98765	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

AGGREGATE FUNCTION

COUNT(*) Function

```
SELECT COUNT(*) FROM chatsystem.users;
```

Result Grid | Filter Rows: | Export:

COUNT(*)
4

MIN Function

```
SELECT MIN(creationdate) FROM chatsystem.groupchat;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

MIN(creationdate)
2002-05-07

MAX Function

```
SELECT MAX(creationdate) FROM chatsystem.groupchat;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MAX(creationdate)
2022-05-02

SUM Function

```
SELECT SUM(age) FROM chatsystem.users;
```

Result Grid | Filter Rows: | Export:

SUM(age)
100

AVG Function

```
SELECT AVG(age) FROM chatsystem.users;
```

Result Grid | Filter Rows: | Export: | Wra

AVG(age)
25.0000

EXIST FUNCTION

```
SELECT username
FROM users
WHERE EXISTS (SELECT userid FROM users WHERE username IN ("Mahrosh"));
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

username
Mahrosh
Haider
Asfar
Sheza

Sub-Queries

```
SELECT * FROM chatsystem.users;
WHERE users=(SELECT userid FROM users WHERE statusmessage = 'feeling dizzy');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

userid
1

Form:

Table1 | Chatsystem | chatsystem_ImportErrors | Chatsystem

Chatsystem

ID	1
Field1	
Field2	username
Field3	statusmessage
Field4	lastseen
Field5	email
Field6	
Field7	contact
Field8	Passwords

Report:

Table1 | Chatsystem | chatsystem_ImportErrors | Chatsystem | Chatsystem

Chatsystem

Saturday, May 18, 2024
7:07:39 PM

ID	Field1	Field2	Field3	Field4	Field5	Field6	Field7
1		username	statusmessage	lastseen	email		contact
2	1	Mahrosh	Feeling dizzy	2024-04-28 10:30	mh@gmail.com	10	98765
3	2	Haider	Feeling blessed	2002-07-14 06:09	haa@gmail.com	16	43210
4	3	Asfar	Feeling blissful	2019-09-05 12:40	er5@gmail.com	29	0334509
5	4	Sheza	Feeling relaxed	2023-07-06 10:12	lak@gmail.com	45	73158

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